

MRR Project - Genes & Breast Cancer

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Description & Statistical Analysis

This study relies on two complementary datasets : a high-dimensional gene expression matrix (**GeneX**) and a detailed clinical dataset (**clinical_data**). Together, they allow us to explore the relationship between molecular profiles and patient outcomes in breast cancer, with a particular focus on **vital status**, which serves as the target variable in our predictive modelling.

Gene Expression Dataset (GeneX)

The **GeneX dataset** contains continuous gene expression measurements for **1,231 patient samples** with ranges between 0 to 20.

Each **row** corresponds to a patient and each **column** corresponds to one of **5,000 genes**, identified by their gene symbols.

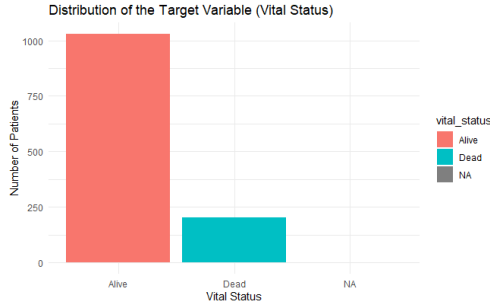
This dataset is used to detect **patterns of co-expression**, identify **groups of biologically similar genes**, and evaluate how these molecular clusters relate to patient survival outcomes.

Clinical Dataset (clinical_data)

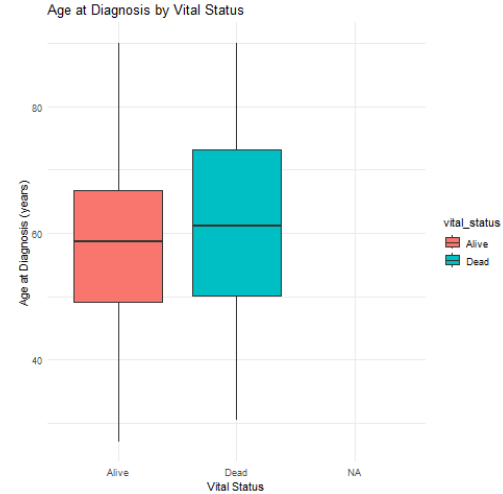
The **clinical_data** dataset contains the clinical and demographic information corresponding to the same **1,231 breast cancer patients** included in GeneX. It consists of **24 variables** covering tumor characteristics, patient attributes, pathological diagnosis, treatment history, and follow-up data.

Summary

Dataset	Number of samples	Data Type	Predictors Number	Structure
GeneX	1231	Continuous numerical	5000	High-dimensional (p » n)
clinical_data	1231	Mixed (categorical, numerical, lists)	24	Tabular, patient-level data



(a) Bar plot - vital status distribution



(b) Boxplot - age at diagnosis by vital status

Figure 1: plots of....

Clustering : grouping genes together

We want to group genes together using k-means algorithm to simplify the visualisation and also we assume that one gene is expressed accordingly with the expression of others particular genes.

We need to transpose our dataset before applying k-means algorithm : each gene should become a row and each patient should become a column.

Before that, however, we must determine **the optimal number of clusters k** : that's how many clusters can be formed without over-segmenting the data or oversimplifying it.

Partitioning around Medoids (PAM & CLARA Methods)

To guide our choice of k , we rely on approaches closely related to k-means, particularly **PAM** (a robust clustering algorithm based on medoids) and its more scalable extension **CLARA**.

Determining the Optimal Number of Clusters for k-means

To select the best value of k , we combine two complementary metrics **WSS** (Within-Cluster Sum of Squares) and **Average silhouette width** which respectfully measure compactness and separation between clusters.

$k = 4$ provides a good compromise : the silhouette width remains relatively high and the elbow plot begins to flatten around this value.

Thus, **four clusters appear to be the optimal grouping for the GeneX genes.**

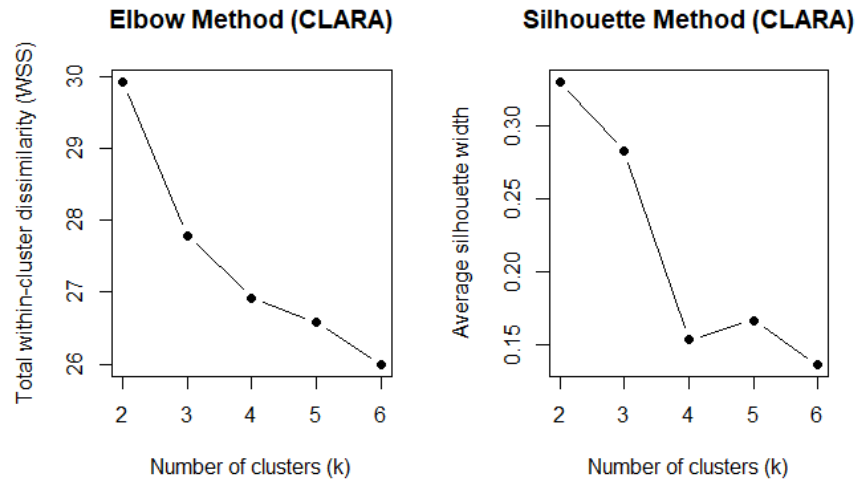


Figure 2: Elbow and silhouette methods graphs to select k